Python

Home Assignments

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Assignment: Student Information and Marks

Write a Python program that accepts the following details from the user using **Scanner**:

1. Student name
2. Student age
3. Student roll number
4. Marks in Java
5. Marks in Python
6. Marks in Database

Calculate and display:

1. Student details
2. Total marks
3. Average marks

No if or loops or functions

Assignment: Employee Salary Calculator

Write a Python program that accepts the following details from the user:

1. Employee name
2. Employee ID
3. Basic salary
4. HRA
5. DA

Calculate and display:

1. Employee details
2. Gross salary = Basic Salary + HRA + DA
3. Annual salary = Gross Salary × 12

No if or loops or functions

Assignment: Shopping Bill Calculator

Write a Python program that accepts:

1. Customer name
2. Product name
3. Product price
4. Quantity

Calculate and display:

1. Total price
2. Discount of 10%
3. Final bill amount after discount

No if or loops or functions

Student Result & Grade Calculator

Create a Python program that accepts a student's marks in **5 subjects**, calculates the total and percentage, and determines:

Pass or Fail

Grade

Division

Scholarship eligibility

Note: Do not use loops, arrays, methods, classes beyond the main class, or switch

Use if statement

Requirements

The student enters:

Student name

Java marks

Python marks

Database marks

Web Technology marks

Aptitude marks

Each subject is out of 100.

Condition 1 - Validate marks

If any mark is:

Less than 0 ⟶ Invalid marks

Greater than 100 ⟶ Invalid marks



Condition 2 - Pass/Fail

A student passes only if **each subject is at least 40.**

Otherwise:

Result: FAIL

Reason: One or more subjects below 40

Condition 3 - Grade

Percentage	Grade
90–100	A+
80–89	A
70–79	B
60–69	C
50–59	D
40–49	E

Condition 4 - Division

Percentage	Division
75+	Distinction
60–74	First Class
50–59	Second Class
40–49	Third Class

Condition 5 - Scholarship

Display:
Scholarship: Eligible
only when:
percentage >= 85 AND every subject >= 75
Otherwise:
Scholarship: Not Eligible

Sample Output

```
===== STUDENT RESULT SYSTEM =====
Enter student name: Rahul
Enter Java marks: 85
Enter Python marks: 78
Enter Database marks: 92
Enter Web Technology marks: 88
Enter Aptitude marks: 81
===== RESULT =====
Student Name : Rahul
Total        : 424 / 500
Percentage   : 84.8%
Result       : PASS
Grade        : A
Division     : First Class
Scholarship  : Not Eligible
```